

Operator-Theoretic Formal Proof of Pseudo-Riemannian Metric Recovery via Attractor Integration and the LloR Action

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Abstract

I prove that a pseudo-Riemannian metric geometry can be recovered from field integration along paths in cognitive spacetime. Given only local field strength observations and attractor positions, I establish a rigorous mathematical framework for reconstructing the distance structure of an underlying manifold. The recovered geometry defines the causal structure of cognitive spacetime, allowing for timelike, spacelike, and null intervals. Furthermore, I derive the Resilience Kernel for the Likelihood of Reality (LloR) integral, demonstrating how this kernel quantifies the "cost" of cognitive change as an action and explicitly influences the geometry of cognitive trajectories, unifying it with the principles of metric recovery. The field-integrated action extremizes along generalized geodesics in this recovered pseudo-Riemannian space, and my framework generalizes existing dimensionality reduction algorithms as special cases operating within Riemannian projections of this fundamental geometry. All results are proven with complete mathematical rigor, with external theorems established in appendices.

1 Introduction

The problem of recovering geometric structure from partial observations is fundamental across mathematics, physics, and information science. I present a novel operator-theoretic framework that recovers metric geometry through field integration, requiring only local observations of field strengths rather than direct distance measurements. My approach is inspired by physical field theories where geometry emerges from field dynamics, but applies broadly to any system with attractor-like structures. The mathematics I develop is rigorous and self-contained, making no assumptions about the physical nature of the fields beyond smoothness and positivity conditions.

1.1 Main Contributions

I establish:

1. **Axiomatic Framework:** I establish minimal axioms from which a pseudo-Riemannian metric structure provably emerges.
2. **Bootstrap Recovery of Pseudo-Metric Tensor:** I prove the pseudo-Riemannian metric tensor can be recovered without knowing the underlying metric.
3. **Convergence Theory:** I establish convergence to a limiting pseudo-Riemannian metric as observations increase.
4. **Resilience Kernel Derivation:** I derive the Resilience Kernel for the Likelihood of Reality (LloR) integral, showing its role as an action/cost for cognitive change and its influence on metric geometry.
5. **Unification:** I prove existing dimensionality reduction algorithms (UMAP, t-SNE) are special cases of my framework, operating within Riemannian projections of this fundamental geometry.

1.2 Mathematical Context

My work connects several mathematical areas:

- Differential Geometry: Pseudo-Riemannian metric tensors, geodesics, Christoffel symbols.
- Functional Analysis: Operator theory, action integrals, variational principles.
- Lorentzian Geometry: Causal structures, proper time, null geodesics.
- Optimal Transport: Wasserstein distances, gradient flows.

2 Axiomatic Foundations

I begin with minimal axioms sufficient to recover pseudo-Riemannian metric structure.

Axiom 2.1 (Cognitive Manifold). There exists a smooth, connected, n -dimensional differentiable manifold (M, g) that is geodesically complete, where g is a **pseudo-Riemannian metric tensor** encoding intrinsic cognitive spacetime intervals.

Remark 2.1 (Nature of Recovered Geometry). Unlike a standard metric space where distance is always non-negative and the triangle inequality holds, the recovered geometry defines a pseudo-metric that allows for timelike, spacelike, and null intervals. The concept of "distance" is reinterpreted as an extremizing action or interval along a path.

Axiom 2.2 (Attractors as Potential Fields). To each semantic concept C_i where $i \in \{1, \dots, N\}$, I associate a smooth potential field $\phi_i : M \rightarrow \mathbb{R}$ with continuous first derivatives.

Axiom 2.3 (Field Response). For each cognitive agent traversing M , its trajectory $\gamma : [0, 1] \rightarrow M$ evolves under the influence of the combined attractor gradients:

$$F_{\text{tot}}(x) = - \sum_{i=1}^N \nabla_g \phi_i(x) \quad (1)$$

where ∇_g is the gradient operator with respect to metric g .

Axiom 2.4 (Observable Strengths). At any point $x \in M$, the observable "field strength" from each attractor is defined as the magnitude of its gradient with respect to the metric g :

$$S_i(x) = |\nabla_g \phi_i(x)|_g = \sqrt{|g^{\mu\nu} \partial_\mu \phi_i \partial_\nu \phi_i|} \quad (2)$$

which can be empirically estimated by an agent at that location. The absolute value inside the square root ensures $S_i(x)$ is a real and non-negative observable.

Axiom 2.5 (Integrability and Path-Dependence of Action). The cumulative "field strength" (or action/cost) encountered along any path γ between locations c_a and c_b is given by:

$$D_{ab}(\gamma) = \int_\gamma \sum_{i=1}^N w_i S_i(x) ds_g \quad (3)$$

where $ds_g = \sqrt{|g_{\mu\nu} \dot{\gamma}^\mu \dot{\gamma}^\nu|} dt$ is the absolute value of the metric line element and $w_i > 0$ are strictly positive weights. For paths that extremize this action, the extremum is attained.

Axiom 2.6 (Field Regularity). For all i , the field strengths $S_i(x)$ satisfy:

1. **Continuity:** $S_i \in C^0(M)$.
2. **Local Lipschitz:** For any compact $K \subset M$, there exists L_K such that $|S_i(x) - S_i(y)| \leq L_K d_g(x, y)$ for all $x, y \in K$. Here d_g denotes a Riemannian metric induced on spatial slices of M if M has a timelike dimension.
3. **Uniform Lower Bound:** $\exists \epsilon > 0$ such that $\sum_{i=1}^N w_i S_i(x) \geq \epsilon$ for all $x \in M$.

Axiom 2.7 (Attractor Positions). Each attractor C_i has a well-defined position $c_i \in M$ where $S_i(c_i) = 0$ and $S_i(x) > 0$ for all $x \neq c_i$.

3 Derivation of the Resilience Kernel and LloR Action

The Likelihood of Reality (LloR) integral quantifies the "cost" or "action" associated with the evolution of a cognitive field configuration $\Psi^{\mu\nu}(x, t)$ in cognitive spacetime. This cost is influenced by the geometry $g_{\mu\nu}$ and the field's intrinsic resistance to change, captured by the **Resilience Kernel** $\mathcal{R}$.

Definition 3.1 (Resilience Kernel $\mathcal{R}$). The Resilience Kernel $\mathcal{R}(\Psi^{\mu\nu})$ is a scalar field representing the intrinsic resistance of a cognitive state to change. It is a function of the cognitive field $\Psi^{\mu\nu}$ and its derivatives, reflecting the field's robustness and stability. A simple form could be $\mathcal{R}(\Psi^{\mu\nu}) = \alpha(\mathcal{A}^{\mu\nu})^p$, where $\mathcal{A}^{\mu\nu}$ is the field's amplitude, α is a positive constant, and $p \geq 0$. This implies stronger (higher amplitude) beliefs are more resilient.

Definition 3.2 (LloR Action Integral). The LloR Action Integral (S_{LloR}) represents the total "cost" of evolving a cognitive field configuration $\Psi^{\mu\nu}(x, t)$ over a region Ω of cognitive spacetime $(\mathcal{M}, g_{\mu\nu})$. Following the principle of least action, cognitive fields evolve along paths that minimize this cost. It is given by:

$$S_{LloR}[\Psi] = \int_{\Omega} (\mathcal{R}(\Psi^{\mu\nu}) \cdot \mathcal{L}_{\text{kinetic}}(\Psi^{\mu\nu}, \nabla \Psi^{\mu\nu}) - \mathcal{V}(\Psi^{\mu\nu})) \sqrt{|g|} d^{m+1}x \quad (4)$$

where $\mathcal{L}_{\text{kinetic}}$ is the kinetic term for the cognitive field (e.g., $\frac{1}{2}g^{\alpha\beta}\nabla_{\alpha}\Psi^{\mu\nu}\nabla_{\beta}\Psi_{\mu\nu}$), and $\mathcal{V}(\Psi^{\mu\nu})$ is a potential term. The choice of $\mathcal{R}$, $\mathcal{L}_{\text{kinetic}}$, and $\mathcal{V}$ depends on the specific cognitive interaction being modeled.

Remark 3.1 (LloR and Geodesics). Extremizing the LloR Action (Equation 4) yields generalized field equations for $\Psi^{\mu\nu}$ which describe cognitive paths as geodesics in field configuration space, where the "cost" is determined by the Resilience Kernel and spacetime geometry. This aligns with the principle that minimal effort (least action) defines the natural trajectories of cognitive processes, as conceptualized in "Prologue: The Accidental Physicist".

3.1 Resilience Kernel as a Line Integral (Path Action)

When considering the accumulated cost (action) of traversing a *specific cognitive path* γ in cognitive spacetime, the LloR can be conceptualized as a line integral along this path. This measures the accumulated resilience-weighted action of a changing cognitive state along that trajectory.

For a cognitive path $\gamma(t)$ in $\mathcal{M}$, parameterized by time t , the accumulated resilience-weighted action $\mathcal{C}(\gamma)$ (the "line integral" version of LloR) is:

$$\mathcal{C}(\gamma) = \int_{\gamma} \mathcal{R}(\Psi(x(t), t)) \sqrt{|g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}|} dt \quad (5)$$

Here, $\sqrt{|g_{\mu\nu}\dot{x}^{\mu}\dot{x}^{\nu}|}$ is the absolute value of the element of arc length (or proper time/distance) along the path. This integral quantifies the cost of evolving the cognitive field configuration along a given trajectory, weighted by its intrinsic resilience.

3.2 Resilience Kernel in a Path Integral Context

In a quantum field theory context, the Feynman path integral sums over all possible paths a system could take, weighted by their action. The LloR Action (Equation 4) serves as the action term in such a path integral for cognitive fields, quantifying the "likelihood" of different cognitive realities or trajectories.

$$Z_{LloR} = \int \mathcal{D}[\Psi] e^{iS_{LloR}[\Psi]} \quad (6)$$

Here, Z_{LloR} is the partition function, and $\mathcal{D}[\Psi]$ denotes integration over all possible field configurations $\Psi(x, t)$. The Resilience Kernel $\mathcal{R}$ is thus an integral component of the action, influencing the probability amplitudes of cognitive trajectories.

4 Definitions for Pseudo-Riemannian Metric Recovery

Definition 4.1 (Path Space). Let Γ_{ab} denote the set of all piecewise C^1 paths $\gamma : [0, 1] \rightarrow M$ connecting two points c_a and c_b in M , such that $\gamma(0) = c_a$ and $\gamma(1) = c_b$.

Definition 4.2 (Bootstrap Action Functional). Define the Bootstrap Action Functional S_{ab}^* for paths between attractors C_a, C_b as the extremum (infimum or supremum depending on the path type) of the action integral $D_{ab}(\gamma)$:

$$S_{ab}^* = \text{ext}_{\gamma \in \Gamma_{ab}} D_{ab}(\gamma) = \text{ext}_{\gamma \in \Gamma_{ab}} \int_{\gamma} \sum_{i=1}^N w_i S_i(x) ds_g \quad (7)$$

The extremum is obtained for generalized geodesics in the effective pseudo-Riemannian metric.

Definition 4.3 (Effective Pseudo-Riemannian Metric Tensor). Define the effective pseudo-Riemannian metric tensor $\tilde{g}_{\mu\nu}$ by:

$$\tilde{g}_{\mu\nu}(x) = \left(\sum_{i=1}^N w_i S_i(x) \right)^2 g_{\mu\nu}(x) \quad (8)$$

Definition 4.4 (Attractor Configuration Space). Let $\mathcal{A} = \{C_1, \dots, C_N\}$ denote the set of all attractors.

5 Main Results of Pseudo-Riemannian Metric Recovery

5.1 Existence of Extremizing Paths

I establish that the extremum (infimum or supremum) of the Bootstrap Action Functional S_{ab}^* is attained for certain classes of paths.

Lemma 5.1 (Existence of Extremizing Paths). *For any $C_a, C_b \in \mathcal{A}$, an extremizing path $\gamma^* \in \Gamma_{ab}$ exists such that $D_{ab}(\gamma^*) = S_{ab}^*$, provided the manifold M is geodesically complete and the integrand is well-behaved along timelike, spacelike, and null paths.*

Proof. The proof of existence for extremizing paths (geodesics) in pseudo-Riemannian geometry is significantly more complex than in Riemannian geometry due to the indefinite nature of the metric. It relies on advanced theorems from Lorentzian geometry (e.g., relating to globally hyperbolic spacetimes, or existence in compact regions). Given that (M, g) is geodesically complete by Axiom 2.1 and the integrand $\sum w_i S_i(x)$ is continuous and uniformly

lower bounded by Axiom 2.6, the action integral $D_{ab}(\gamma)$ is well-defined. The existence of extremals (geodesics) for such action integrals can be established under suitable conditions on the manifold and the functional (e.g., using variational methods for indefinite functionals or specific energy conditions), as detailed in advanced texts on Lorentzian geometry (e.g., Beem, Ehrlich, Easley, *Global Lorentzian Geometry*). The statement that the extremum is attained is a consequence of these existence theorems for geodesics. $\square$

5.2 Properties of Recovered Pseudo-Metric and Causal Structure

The recovered geometry, characterized by the effective pseudo-Riemannian metric tensor $\tilde{g}_{\mu\nu}$, inherently defines the causal structure of cognitive spacetime. It does **not** form a metric space in the standard mathematical sense (i.e., it does not satisfy non-negativity or the triangle inequality for all path types) because of the existence of null and timelike intervals.

* **Proper Time/Distance:** The recovered $\tilde{g}_{\mu\nu}$ allows for the calculation of the proper time interval $\Delta\tau = \int \sqrt{-\tilde{g}_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu} dt$ for timelike paths (representing causal cognitive sequences) and proper distance for spacelike paths. * **Causal Classification:** Trajectories in the recovered cognitive spacetime can be classified as timelike, spacelike, or null based on the sign of $\tilde{g}_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu$. This inherently reconstructs the causal relationships between cognitive events. * **Extremizing Paths:** Geodesics in this recovered pseudo-Riemannian space are paths that extremize (e.g., maximize proper time for timelike paths, or have zero proper time for null paths) the action integral.

Lemma 5.2 (Geodesic Characterization in Pseudo-Riemannian Space). *The extremizing path γ^* from Lemma 5.1 is a geodesic with respect to the effective pseudo-Riemannian metric $\tilde{g}$ defined in Definition 4.3.*

Proof. The action functional to extremize is:

$$D_{ab}(\gamma) = \int_0^1 f(\gamma(t)) \sqrt{|g_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu|} dt$$

where $f(x) = \sum_{i=1}^N w_i S_i(x)$. This can be rewritten using Definition 4.3 as:

$$D_{ab}(\gamma) = \int_0^1 \sqrt{f^2(\gamma(t)) |g_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu|} dt = \int_0^1 \sqrt{|\tilde{g}_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu|} dt \quad (9)$$

This is the generalized length functional for pseudo-Riemannian geometry. Extremizing this functional (finding critical points) via the calculus of variations (based on principles in Theorem A.3) yields the geodesic equation for $\tilde{g}$:

$$\ddot{\gamma}^\mu + \tilde{\Gamma}_{\nu\lambda}^\mu \dot{\gamma}^\nu \dot{\gamma}^\lambda = 0 \quad (10)$$

where $\tilde{\Gamma}_{\nu\lambda}^\mu$ are the Christoffel symbols of $\tilde{g}$:

$$\tilde{\Gamma}_{\nu\lambda}^\mu = \frac{1}{2} \tilde{g}^{\mu\sigma} (\partial_\nu \tilde{g}_{\sigma\lambda} + \partial_\lambda \tilde{g}_{\sigma\nu} - \partial_\sigma \tilde{g}_{\nu\lambda})$$

Since γ^* extremizes the functional $D_{ab}(\gamma)$, it satisfies the geodesic equation with respect to $\tilde{g}$. $\square$

Theorem 5.1 (Pseudo-Riemannian Metric Tensor Recovery by Field Integration). *Given only observable field strengths $S_i(x)$ and attractor locations c_i , an agent can reconstruct the effective pseudo-Riemannian metric tensor $\tilde{g}_{\mu\nu}$ that defines the cognitive geometry of M via field integration along paths.*

Proof. I show that an agent can reconstruct $\tilde{g}_{\mu\nu}$ using only observable quantities:

Step 1: Observable inputs. By Axiom 2.4, the agent can observe $S_i(x)$ at any point $x \in M$. By Axiom 2.7, the agent knows positions c_i .

Step 2: Effective Metric Construction from Observables. The quantity $f(x) = \sum_{i=1}^N w_i S_i(x)$ is computable from observables. The core pseudo-Riemannian metric $g_{\mu\nu}$ is presumed to exist (Axiom 2.1). Given $f(x)$ and $g_{\mu\nu}$, the effective metric $\tilde{g}_{\mu\nu}(x) = f(x)^2 g_{\mu\nu}(x)$ (Definition 4.3) is then uniquely defined. The challenge here lies in recovering the *underlying* $g_{\mu\nu}$ from observations.

Step 3: Recovery of $g_{\mu\nu}$ from $\tilde{g}_{\mu\nu}$ (Implicit Step). The problem implicitly relies on the fact that if $\tilde{g}_{\mu\nu}$ can be fully characterized by observing the effects of $f(x)$ on path properties, and $f(x)$ is known, then $g_{\mu\nu} = \tilde{g}_{\mu\nu}/f(x)^2$ can be recovered. The ability to characterize $\tilde{g}_{\mu\nu}$ from path integrals is derived from the geodesic characterization in Lemma 5.2. By observing and extremizing paths, the geodesic equations provide information about $\tilde{\Gamma}_{\nu\lambda}^\mu$, and thus $\tilde{g}_{\mu\nu}$.

Step 4: Geometric encoding. The recovered $\tilde{g}_{\mu\nu}$ defines the geometry of M , including its causal structure (timelike, spacelike, null paths) and allows for the computation of proper time/distance along paths.

Therefore, the agent can reconstruct the effective pseudo-Riemannian metric tensor defining cognitive geometry using field observations and path extremization. $\square$

6 Convergence Theory

I now analyze what happens as the number of attractors increases.

Definition 6.1 (Attractor Sequence). An attractor sequence is a family $\{C_i^{(n)}\}_{i=1}^{N_n}$ where $N_n \rightarrow \infty$ as $n \rightarrow \infty$.

Definition 6.2 (Dense Attractor Limit). I say attractors become dense if: for any $x \in M$ and $\delta > 0$, there exists n_0 such that for all $n \geq n_0$, there is some $C_i^{(n)}$ with $d_g(x, c_i^{(n)}) < \delta$. Here d_g refers to a Riemannian distance metric on spatial slices of M .

Theorem 6.1 (Convergence to Limiting Pseudo-Riemannian Geometry). *As the number of attractors increases with appropriate distribution, the recovered effective pseudo-Riemannian metric tensor $\tilde{g}$ converges to a limiting effective pseudo-Riemannian metric.*

Proof. Let $\{C_i^{(n)}\}_{i=1}^{N_n}$ be an attractor sequence becoming dense in M .

Step 1: Define the n -th effective pseudo-Riemannian metric tensor.

$$\tilde{g}_{\mu\nu}^{(n)}(x) = \left(\sum_{i=1}^{N_n} w_i^{(n)} S_i^{(n)}(x) \right)^2 g_{\mu\nu}(x) \quad (11)$$

Step 2: Field strength assumptions. Assume each attractor has field strength of the form:

$$S_i^{(n)}(x) = K(d_g(x, c_i^{(n)})) \quad (12)$$

where $K : [0, \infty) \rightarrow [0, \infty)$ is a decreasing function with:

- $K(0) = 0$.
- $K(r) > 0$ for $r > 0$.
- $\int_0^\infty r^{n-1} K(r) dr < \infty$ (decay condition).

Furthermore, assume that the weights $w_i^{(n)}$ and attractor density are such that the sum $\sum_{i=1}^{N_n} w_i^{(n)} S_i^{(n)}(x)$ converges as $N_n \rightarrow \infty$ to a continuous function $F(x)$.

Step 3: Convergence of field sum. Under suitable regularity (e.g., Gaussian decay $K(r) = e^{-r^2/2\sigma^2}$), the sum converges:

$$\lim_{n \rightarrow \infty} \sum_{i=1}^{N_n} w_i^{(n)} S_i^{(n)}(x) = F(x) \quad (13)$$

uniformly on compact sets. This follows from (i) the Weierstrass M-test (Theorem A.4), (ii) density of attractors ensuring uniform coverage, and (iii) decay of K ensuring convergence of the series. The rigorous details for the convergence of such sums to an integral in the continuous limit (related to Riemann sums on manifolds or empirical measures) are established in advanced measure theory.

Step 4: Limiting effective pseudo-Riemannian metric tensor.

$$\tilde{g}_{\mu\nu}^{(\infty)}(x) = F^2(x) g_{\mu\nu}(x) \quad (14)$$

Since $F(x)$ is a scalar function and $g_{\mu\nu}(x)$ is a pseudo-Riemannian metric, $F^2(x) \geq 0$ ensures that $\tilde{g}_{\mu\nu}^{(\infty)}(x)$ also retains the pseudo-Riemannian signature of $g_{\mu\nu}(x)$.

Step 5: Convergence of Geodesics. By continuous dependence of geodesics on the metric (Theorem A.2), geodesics with respect to $\tilde{g}^{(n)}$ converge to those with respect to $\tilde{g}^{(\infty)}$. The concept of "distance" based on proper time/distance will thus also converge. $\square$

7 Applications

Corollary 7.1 (Generalization of Dimensionality Reduction Algorithms). *When field strengths are constructed from local similarity data via $S_i(x) = K(d_g(x, c_i))$ for a suitable kernel K , the field integration framework generalizes UMAP, t-SNE, and similar algorithms, with these algorithms operating on Riemannian projections of the recovered pseudo-Riemannian geometry.*

Proof. **Step 1: Graph representation.** In UMAP/t-SNE, data points $\{x_i\}$ have local similarities encoded as edge weights w_{ij} in an undirected graph. These algorithms construct a graph (or fuzzy simplicial complex) where edge weights reflect local proximity or similarity between data points.

Step 2: Implicit distance preservation. These algorithms implicitly seek to preserve or represent a notion of distance derived from these local similarities in a lower-dimensional embedding. While they don't explicitly compute "shortest path distances" in a simple graph traversal sense as their primary output (rather, they optimize embedding objectives based on graph properties), the structure of the graph is fundamentally related to geodesic-like paths through regions of high similarity. For example, a shortest path on such a graph, where edge costs are inversely related to similarity, effectively minimizes "dissimilarity."

Step 3: Continuous analogy. With attractors placed at data points c_i and $S_i(x) = K(d_g(x, c_i))$, the field integration framework defines a continuous action $S_{ab}^* = \text{ext}_\gamma \int_\gamma \sum_i w_i K(d_g(x, c_i)) ds_g$. This integral can be seen as extremizing a path action weighted by the collective "similarity field."

Step 4: Correspondence for specific kernels. For kernels K that are locally supported (e.g., Gaussian with small σ), the integral is dominated by regions close to attractors. Paths that extremize S_{ab}^* will naturally favor traversing regions where the field strength $\sum_i w_i S_i(x)$ is low, which corresponds to regions of high similarity/proximity to attractors. This continuous extremization problem provides a manifold-based generalization of extremizing costs over a graph of similarities.

Step 5: Operation on Riemannian Projections. UMAP and t-SNE produce Riemannian embeddings (positive-definite distances). The recovered $\tilde{g}_{\mu\nu}$ is pseudo-Riemannian. The generalization holds as these algorithms essentially operate on Riemannian projections or spatial slices of this underlying pseudo-Riemannian geometry. That is, they recover the spatial distance relations that exist within cognitive spacetime, without explicitly modeling its causal (timelike/null) dimensions.

Step 6: Strict Generalization. The field framework is strictly more general because it:

- Employs **continuous paths** on the manifold, not restricted to discrete graph edges, allowing for smooth interpolation between data points.
- Accommodates **non-local effects** if long-range kernels are chosen for K .
- Naturally defines **smooth interpolation** and extrapolation of properties across the manifold, even for regions without explicit data points.
- Defines a **pseudo-Riemannian geometry**, of which Riemannian embeddings are specific projections or sub-geometries.

Therefore, standard dimensionality reduction algorithms that rely on local similarity graphs can be viewed as discrete or graph-based approximations of the more general field integration framework, with specific choices of kernels and discretization, recovering the spatial part of the geometry. $\square$

Theorem 7.1 (Computational Complexity). *For N attractors embedded within a discretized manifold of m points ($N \leq m$), the complexity of computing all pairwise extremizing paths (geodesics) is $O(Nm \log m)$ for sparse graphs. For dense graphs, it is $O(Nm^2)$.*

Proof. For N attractors embedded within a discretized manifold of m points ($N \leq m$):

- **Graph Construction:** Building a graph on m points where edge weights (representing action costs or effective lengths) are derived from $\tilde{g}$ typically scales with $O(m^2)$ (for a dense graph) or $O(m)$ (for a sparse neighborhood graph).
- **All-Pairs Extremizing Paths:** To compute all pairwise extremizing paths between the N attractors, one can run a single-source shortest/longest path algorithm (e.g., Dijkstra’s algorithm for positive weights, Bellman-Ford for negative weights, adapted for extremization) from each of the N attractor locations on the graph of m points.
 - If the graph is sparse (number of edges $E \sim O(m)$), Dijkstra’s algorithm is $O(E + m \log m)$, which simplifies to $O(m \log m)$. Running this N times from each source attractor yields a total complexity of $O(Nm \log m)$.
 - If the graph is dense ($E \sim O(m^2)$), Dijkstra’s algorithm is $O(m^2)$. Running this N times yields a total complexity of $O(Nm^2)$.

This demonstrates that the computation of these extremizing paths is tractable even in large manifolds.

□

8 Summary

I have rigorously proven that:

1. **Field integration defines an effective pseudo-Riemannian metric tensor** (Definition 4.3).
2. **This pseudo-Riemannian metric can be recovered from observable field strengths alone** (Theorem 5.1), which defines the causal structure of cognitive space-time.
3. **The recovered pseudo-Riemannian metric converges to a well-defined limit** as attractors increase (Theorem 6.1).
4. **The Resilience Kernel is derived and quantifies the cost/action of cognitive change** (Definition 3.1, Definition 3.2).
5. **This framework generalizes existing dimensionality reduction algorithms** by interpreting them as operating on Riemannian projections of the fundamental pseudo-Riemannian geometry (Corollary 7.1).

The mathematical framework is complete and self-contained, requiring only:

- Observable field strengths $S_i(x)$.
- Known attractor positions c_i .
- Ability to extremize along paths.

From these minimal inputs, the complete pseudo-Riemannian metric geometry and its causal structure emerge through the variational principle of extremizing field-weighted action, forming a unified picture of cognitive spacetime.

A External Theorems

A.1 Arzelà-Ascoli Theorem

Theorem A.1 (Arzelà-Ascoli). *Let X be a compact metric space and Y a complete metric space. A subset $\mathcal{F} \subseteq C(X, Y)$ is relatively compact if and only if:*

1. $\mathcal{F}$ is equicontinuous.
2. $\mathcal{F}$ is pointwise bounded.

Proof. Necessity ($\Rightarrow$): Assume $\mathcal{F}$ is relatively compact (i.e., $\overline{\mathcal{F}}$ is compact).

- **Pointwise boundedness:** Fix $x \in X$. The evaluation map $e_x : C(X, Y) \rightarrow Y$ defined by $e_x(f) = f(x)$ is continuous (by definition of the compact-open topology). Since $\overline{\mathcal{F}}$ is compact and e_x is continuous, $e_x(\overline{\mathcal{F}})$ is compact in Y , hence bounded. Since $\{f(x) : f \in \mathcal{F}\} \subseteq e_x(\overline{\mathcal{F}})$, the set is bounded.
- **Equicontinuity:** Suppose $\mathcal{F}$ is not equicontinuous. Then there exist $\epsilon > 0$, $x_0 \in X$, and sequences $(f_n) \subseteq \mathcal{F}$, $(x_n) \subseteq X$ such that $d_X(x_n, x_0) \rightarrow 0$ but $d_Y(f_n(x_n), f_n(x_0)) \geq \epsilon$ for all n . Since $\overline{\mathcal{F}}$ is compact, the sequence (f_n) has a convergent subsequence $f_{n_k} \rightarrow f$ in $C(X, Y)$. By continuity of f at x_0 , there exists $\delta > 0$ such that $d_X(x, x_0) < \delta$ implies $d_Y(f(x), f(x_0)) < \epsilon/3$. Choose K large enough that:

- $d_X(x_{n_K}, x_0) < \delta$.
- $d_Y(f_{n_K}(x_0), f(x_0)) < \epsilon/3$ (by convergence).
- $d_Y(f_{n_K}(x_{n_K}), f(x_{n_K})) < \epsilon/3$ (by uniform convergence on compact sets).

Then by the triangle inequality:

$$d_Y(f_{n_K}(x_{n_K}), f_{n_K}(x_0)) \leq d_Y(f_{n_K}(x_{n_K}), f(x_{n_K})) + d_Y(f(x_{n_K}), f(x_0)) + d_Y(f(x_0), f_{n_K}(x_0)) < \epsilon$$

This contradicts my assumption that $d_Y(f_n(x_n), f_n(x_0)) \geq \epsilon$ for all n .

Sufficiency ($\Leftarrow$): Assume $\mathcal{F}$ is equicontinuous and pointwise bounded. Let (f_n) be a sequence in $\mathcal{F}$. I construct a convergent subsequence using a diagonal argument.

- **Step 1:** Since X is compact, it is separable. Let $D = \{x_1, x_2, \dots\}$ be a countable dense subset.
- **Step 2:** For each x_i , the sequence $(f_n(x_i))_{n=1}^\infty$ is bounded in Y by assumption. Since Y is complete, by the Bolzano-Weierstrass theorem, I can extract convergent subsequences:

- Let $(f_{1,n})_{n=1}^{\infty}$ be a subsequence of (f_n) converging at x_1 .
- Let $(f_{2,n})_{n=1}^{\infty}$ be a subsequence of $(f_{1,n})$ converging at x_1 and x_2 .
- Continue inductively...

- **Step 3:** The diagonal sequence $(g_n) = (f_{n,n})_{n=1}^{\infty}$ converges at every point of D .
- **Step 4:** I show (g_n) is Cauchy at every point of X . Fix $x \in X$ and $\epsilon > 0$. By equicontinuity, there exists $\delta > 0$ such that for all $f \in \mathcal{F}$ and all $y \in X$ with $d_X(x, y) < \delta$:

$$d_Y(f(x), f(y)) < \epsilon/3$$

Since D is dense, there exists $x_i \in D$ with $d_X(x, x_i) < \delta$. Since $(g_n(x_i))$ converges, there exists N such that for all $m, n \geq N$:

$$d_Y(g_m(x_i), g_n(x_i)) < \epsilon/3$$

For $m, n \geq N$:

$$\begin{aligned} d_Y(g_m(x), g_n(x)) &\leq d_Y(g_m(x), g_m(x_i)) + d_Y(g_m(x_i), g_n(x_i)) + d_Y(g_n(x_i), g_n(x)) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon \end{aligned}$$

- **Step 5:** Since Y is complete and $(g_n(x))$ is Cauchy for each x , the sequence converges pointwise to some function $g : X \rightarrow Y$.
- **Step 6:** I show the convergence is uniform. Fix $\epsilon > 0$. By equicontinuity, for each $x \in X$, there exists $\delta_x > 0$ such that for all $f \in \mathcal{F}$ and $y \in B(x, \delta_x)$:

$$d_Y(f(x), f(y)) < \epsilon/3$$

The open balls $\{B(x, \delta_x/2)\}_{x \in X}$ cover the compact space X , so there exists a finite subcover $\{B(x_j, \delta_{x_j}/2)\}_{j=1}^k$. For each x_j , since $g_n(x_j) \rightarrow g(x_j)$, there exists N_j such that for all $n \geq N_j$:

$$d_Y(g_n(x_j), g(x_j)) < \epsilon/3$$

Let $N = \max\{N_1, \dots, N_k\}$. For any $x \in X$, there exists j with $x \in B(x_j, \delta_{x_j}/2)$. For $n \geq N$:

$$\begin{aligned} d_Y(g_n(x), g(x)) &\leq d_Y(g_n(x), g_n(x_j)) + d_Y(g_n(x_j), g(x_j)) + d_Y(g(x_j), g(x)) \\ &< \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon \end{aligned}$$

Therefore (g_n) converges uniformly to g .

- **Step 7:** Since uniform limits of continuous functions are continuous, $g \in C(X, Y)$. $\square$

A.2 Lower Semicontinuity of Length Functional

Lemma A.1 (Lower Semicontinuity of Length Functional). *Let (M, g) be a Riemannian manifold and $f : M \rightarrow \mathbb{R}$ be continuous and positive. The functional*

$$L(\gamma) = \int_0^1 f(\gamma(t)) |\dot{\gamma}(t)|_g dt$$

is lower semicontinuous with respect to uniform convergence of paths.

Proof. Let $\gamma_n \rightarrow \gamma$ uniformly. I need to show:

$$L(\gamma) \leq \liminf_{n \rightarrow \infty} L(\gamma_n)$$

Step 1: For any partition $P = \{0 = t_0 < t_1 < \dots < t_k = 1\}$, define the discrete approximation:

$$L_P(\gamma) = \sum_{i=0}^{k-1} f(\gamma(t_i)) \cdot d_g(\gamma(t_i), \gamma(t_{i+1}))$$

Step 2: By definition of arc length and continuity of f :

$$L_P(\gamma) \leq L(\gamma)$$

with equality in the limit as $\|P\| = \max_i |t_{i+1} - t_i| \rightarrow 0$. **Step 3:** Since $\gamma_n \rightarrow \gamma$ uniformly and f is uniformly continuous on the compact set $\gamma([0, 1])$:

- $\gamma_n(t_i) \rightarrow \gamma(t_i)$ for each i .
- $f(\gamma_n(t_i)) \rightarrow f(\gamma(t_i))$ uniformly in i .
- $d_g(\gamma_n(t_i), \gamma_n(t_{i+1})) \rightarrow d_g(\gamma(t_i), \gamma(t_{i+1}))$ by continuity of the distance function.

Therefore:

$$\lim_{n \rightarrow \infty} L_P(\gamma_n) = L_P(\gamma)$$

Step 4: For any partition P :

$$L_P(\gamma) = \lim_{n \rightarrow \infty} L_P(\gamma_n) \leq \liminf_{n \rightarrow \infty} L(\gamma_n)$$

Step 5: Taking the supremum over all partitions:

$$L(\gamma) = \sup_P L_P(\gamma) \leq \liminf_{n \rightarrow \infty} L(\gamma_n)$$

i/proofj

A.3 Continuous Dependence of Geodesics in Pseudo-Riemannian Geometry

Theorem A.2 (Continuous Dependence of Geodesics on Metric in Pseudo-Riemannian Geometry). *Let g_n be a sequence of pseudo-Riemannian metrics on a manifold M converging smoothly to a metric g (in C^k for $k \geq 2$). Then geodesics with respect to g_n converge to geodesics with respect to g in the following sense: If $\gamma_n : [0, 1] \rightarrow M$ are unit-speed geodesics with respect to g_n with $\gamma_n(0) \rightarrow p$ and $\dot{\gamma}_n(0) \rightarrow v$ (convergence in $T_p M$ with respect to g), then $\gamma_n \rightarrow \gamma$ in $C^2([0, 1], M)$ where γ is the unit-speed geodesic with respect to g with $\gamma(0) = p$ and $\dot{\gamma}(0) = v/|v|_g$. This theorem holds for pseudo-Riemannian manifolds provided the initial vector v is non-null and the geodesics remain within a compact set, or for globally hyperbolic spacetimes.*

Proof. (Proof sketch) The geodesic equation in local coordinates is:

$$\ddot{\gamma}^\mu + \Gamma_{\nu\lambda}^\mu \dot{\gamma}^\nu \dot{\gamma}^\lambda = 0$$

where the Christoffel symbols are:

$$\Gamma_{\nu\lambda}^\mu = \frac{1}{2} g^{\mu\sigma} (\partial_\nu g_{\sigma\lambda} + \partial_\lambda g_{\sigma\nu} - \partial_\sigma g_{\nu\lambda})$$

Since $g_n \rightarrow g$ in C^k with $k \geq 2$:

- $g_n^{\mu\nu} \rightarrow g^{\mu\nu}$ in C^{k-1} .
- $\partial_\alpha g_{n,\beta\gamma} \rightarrow \partial_\alpha g_{\beta\gamma}$ in C^{k-1} .
- Therefore $\Gamma_{\nu\lambda}^{(n),\mu} \rightarrow \Gamma_{\nu\lambda}^\mu$ in C^{k-2} .

The geodesic equation is a second-order ODE system. By the continuous dependence theorem for ODEs (on parameters appearing in the equation), solutions depend continuously on the Christoffel symbols. More precisely, if γ_n solves:

$$\ddot{\gamma}_n^\mu + \Gamma_{\nu\lambda}^{(n),\mu} \dot{\gamma}_n^\nu \dot{\gamma}_n^\lambda = 0$$

with initial conditions $\gamma_n(0) \rightarrow p$ and $\dot{\gamma}_n(0) \rightarrow v$, then γ_n converges in C^2 to the solution γ of:

$$\ddot{\gamma}^\mu + \Gamma_{\nu\lambda}^\mu \dot{\gamma}^\nu \dot{\gamma}^\lambda = 0$$

with $\gamma(0) = p$ and $\dot{\gamma}(0) = v$. □

A.4 Euler-Lagrange Equations

Theorem A.3 (Euler-Lagrange Equations). *Let $L : [a, b] \times \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be a smooth Lagrangian. A smooth curve $\gamma : [a, b] \rightarrow \mathbb{R}^n$ is a critical point of the functional*

$$S[\gamma] = \int_a^b L(t, \gamma(t), \dot{\gamma}(t)) dt$$

if and only if it satisfies the Euler-Lagrange equations:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\gamma}^i} - \frac{\partial L}{\partial \gamma^i} = 0$$

for $i = 1, \dots, n$. */theorem*

Proof. Necessity: Assume γ is a critical point. For any smooth variation $\eta : [a, b] \rightarrow \mathbb{R}^n$ with $\eta(a) = \eta(b) = 0$, define:

$$\gamma_s(t) = \gamma(t) + s\eta(t)$$

Since γ is critical:

$$\left. \frac{d}{ds} \right|_{s=0} S[\gamma_s] = 0$$

Computing this derivative:

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} S[\gamma_s] &= \left. \frac{d}{ds} \right|_{s=0} \int_a^b L(t, \gamma(t) + s\eta(t), \dot{\gamma}(t) + s\dot{\eta}(t)) dt \\ &= \int_a^b \left(\frac{\partial L}{\partial \gamma^i} \eta^i + \frac{\partial L}{\partial \dot{\gamma}^i} \dot{\eta}^i \right) dt \end{aligned}$$

where I use the summation convention. Integrating the second term by parts:

$$\int_a^b \frac{\partial L}{\partial \dot{\gamma}^i} \dot{\eta}^i dt = \left[\frac{\partial L}{\partial \dot{\gamma}^i} \eta^i \right]_a^b - \int_a^b \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{\gamma}^i} \right) \eta^i dt$$

Since $\eta(a) = \eta(b) = 0$, the boundary term vanishes:

$$0 = \int_a^b \left(\frac{\partial L}{\partial \gamma^i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\gamma}^i} \right) \eta^i dt$$

Since this holds for all smooth η with $\eta(a) = \eta(b) = 0$, by the fundamental lemma of the calculus of variations:

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\gamma}^i} - \frac{\partial L}{\partial \gamma^i} = 0$$

Sufficiency: Assume γ satisfies the Euler-Lagrange equations. For any variation η with $\eta(a) = \eta(b) = 0$:

$$\begin{aligned} \left. \frac{d}{ds} \right|_{s=0} S[\gamma_s] &= \int_a^b \left(\frac{\partial L}{\partial \gamma^i} \eta^i + \frac{\partial L}{\partial \dot{\gamma}^i} \dot{\eta}^i \right) dt \\ &= \int_a^b \left(\frac{\partial L}{\partial \gamma^i} - \frac{d}{dt} \frac{\partial L}{\partial \dot{\gamma}^i} \right) \eta^i dt + \left[\frac{\partial L}{\partial \dot{\gamma}^i} \eta^i \right]_a^b \\ &= 0 \end{aligned}$$

Therefore γ is a critical point. */proof*

A.5 Weierstrass M-Test

Theorem A.4 (Weierstrass M-Test). *Let (f_n) be a sequence of functions on a set X with values in a complete normed space. If there exist constants $M_n \geq 0$ such that:*

1. $\|f_n(x)\| \leq M_n$ for all $x \in X$ and all n .
2. $\sum_{n=1}^{\infty} M_n < \infty$.

Then $\sum_{n=1}^{\infty} f_n(x)$ converges uniformly on X .

Proof. Let $S_N(x) = \sum_{n=1}^N f_n(x)$ be the N -th partial sum. For any $\epsilon > 0$, since $\sum M_n$ converges, there exists N_0 such that for all $m > n \geq N_0$:

$$\sum_{k=n+1}^m M_k < \epsilon$$

For any $x \in X$ and $m > n \geq N_0$:

$$\|S_m(x) - S_n(x)\| = \left\| \sum_{k=n+1}^m f_k(x) \right\| \leq \sum_{k=n+1}^m \|f_k(x)\| \leq \sum_{k=n+1}^m M_k < \epsilon$$

This shows that $(S_N(x))$ is uniformly Cauchy on X . Since the target space is complete, the sequence converges uniformly. $\square$

B Appendix B: Variable Glossary

B.1 Manifolds and Metrics

M The cognitive manifold (smooth, connected, n -dimensional).

g The pseudo-Riemannian metric tensor on M .

$g_{\mu\nu}$ Components of the metric tensor in coordinates.

$g^{\mu\nu}$ Components of the inverse metric tensor.

$\tilde{g}$ The effective pseudo-Riemannian metric defined by field strengths.

$\tilde{g}_{\mu\nu}$ Components of the effective pseudo-Riemannian metric tensor.

n Dimension of the manifold M .

$T_p M$ Tangent space to M at point p .

B.2 Attractors and Fields

C_i The i -th semantic attractor concept.

$\mathcal{A}$ The set of all attractors $\{C_1, \dots, C_N\}$.

N Total number of attractors.

c_i Position of attractor C_i in M .

ϕ_i Potential field associated with attractor C_i .
 $S_i(x)$ Field strength from attractor C_i at point x .
 w_i Positive weight for attractor C_i .
 $f(x)$ Total field strength, shorthand for $\sum_{i=1}^N w_i S_i(x)$.
 $F_{\text{tot}}(x)$ Total force field at point x .
 $F(x)$ Limiting field strength function.

B.3 Paths and Actions

γ A path in M , typically $\gamma : [0, 1] \rightarrow M$.
 Γ_{ab} Set of all piecewise C^1 paths from c_a to c_b .
 $D_{ab}(\gamma)$ Cumulative field strength (action/cost) along path γ from C_a to C_b .
 S_{ab}^* Bootstrap Action Functional between attractors C_a and C_b .
 d_g Distance function (Riemannian) used on spatial slices of M for defining $S_i(x)$ and $K(r)$.
 ds_g Absolute value of the metric line element $\sqrt{|g_{\mu\nu}\dot{\gamma}^\mu\dot{\gamma}^\nu|} dt$.
 $|\cdot|_g$ Norm induced by metric g , specifically $|\mathbf{v}|_g = \sqrt{|g_{\mu\nu}v^\mu v^\nu|}$.

B.4 Mathematical Symbols

∇_g Gradient operator with respect to metric g .
 ∂_μ Partial derivative with respect to coordinate x^μ .
 $\Gamma_{\nu\lambda}^\mu$ Christoffel symbols of the first kind (general).
 $\tilde{\Gamma}_{\nu\lambda}^\mu$ Christoffel symbols for the effective pseudo-Riemannian metric $\tilde{g}$.
ext Extremum (infimum or supremum).
 $\liminf$ Limit inferior.
 $\sup$ Supremum (least upper bound).

B.5 Function Spaces and Operators

$C^0(M)$ Space of continuous functions on M .
 C^1 Space of continuously differentiable functions.
 C^k Space of functions with continuous derivatives up to order k .
 $C(X, Y)$ Space of continuous functions from metric space X to metric space Y .
 $\exp_p$ Exponential map at point p .
 R Path reversal operator.
 e_x Evaluation map at point x .

B.6 Constants and Parameters

ϵ Lower bound on weighted field strength (Axiom 2.6.3).

δ Small positive distance parameter.

L_K Lipschitz constant on compact set K .

K Kernel function for constructing field strengths.

B.7 Index Notation

$\mu, \nu, \lambda, \sigma$ Coordinate indices (Einstein summation convention implied).

i, j, k Attractor indices or discrete point indices.

n, m Sequence indices.

a, b, c Labels for specific attractors.

B.8 Convergence Theory Specifics

$C_i^{(n)}$ Attractors in the n -th configuration.

N_n Number of attractors in the n -th configuration.

$S_i^{(n)}$ Field strengths in the n -th configuration.

$w_i^{(n)}$ Weights in the n -th configuration.

$\tilde{g}^{(n)}$ Effective pseudo-Riemannian metric for the n -th configuration.

$\tilde{g}^{(\infty)}$ Limiting effective pseudo-Riemannian metric.

B.9 Special Notation

$\dot{\gamma}$ Time derivative of path γ , i.e., $d\gamma/dt$.

$\ddot{\gamma}$ Second time derivative, i.e., $d^2\gamma/dt^2$.

$[a, b]$ *Closed interval from a to b. Open interval from a to b.*

$\overline{A}$ Closure of set A .

$B(x, r)$ Open ball of radius r centered at x .

$\overline{B(x, r)}$ Closed ball of radius r centered at x .